

## Universal Balance Laws

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Universal Balance Laws is a minimal physical update system built from four expressions. These expressions describe how density changes, how speed reacts to that change, how influence weakens with distance, and how tightly the two local quantities are bound together. The system does not assume fields, forces, particles, or any external constructs. It is defined entirely by arithmetic updates that operate directly on the substrate.

The update for density is the simplest part of the system. Density changes only because flow is pushed through it by speed. If  $D$  is the current density,  $S$  is the current speed, and  $D_{\text{flow}}$  is the amount of flow passing through the substrate, then the next density is

$$D_{\text{next}} = D + D_{\text{flow}}S.$$

This expression is the literal mechanism of change. Density increases by exactly the amount of flow multiplied by speed. There is no hidden term or secondary correction. The substrate moves because speed pushes flow, and the amount pushed is  $D_{\text{flow}}S$ . This is the entire local motion of the system.

Speed updates through the same driver. The next speed is

$$S_{\text{next}} = S + kD_{\text{flow}}S.$$

This shows that speed is not an independent quantity. It reacts directly to the same change that moved density. The term  $D_{\text{flow}}S$  appears again, proving that the substrate uses one driver for both updates. The constant  $k$  determines how strongly speed reacts to each unit of density movement. When  $k$  is small, speed barely responds; when  $k$  is large, speed amplifies quickly. The substrate compounds because both density and speed ride on the same flow-through-speed term.

Influence or gravity updates through a separate geometric rule. If  $G$  is the influence between two points and distance is the separation between them, then the next influence is

$$G_{\text{next}} = \frac{G}{\text{distance}^2}$$

This inverse-square weakening is independent of the density–speed coupling. Doubling the distance reduces influence to one quarter; tripling the distance reduces it to one ninth. This geometric cost ensures that the strong local coupling described earlier does not propagate without limit. The substrate moves locally but weakens globally.

The relationship between the two local updates is captured by a single ratio:

$$\frac{\Delta S}{\Delta D} = k.$$

This ratio states that the change in speed is always a constant fraction of the change in density. It is the algebraic proof that the density update and speed update are two expressions of one underlying process. The constant  $k$  is the reaction coefficient of the substrate, determining how tightly the two quantities track each other.

These four expressions — the density update, the speed update, the gravity update, and the coupling ratio — are the complete system. They describe a substrate that moves locally through flow pushed by speed, reacts proportionally through a coupling constant, and weakens globally through geometric distance. Nothing else is required.